A 50-Ω lossless transmission line is connected to a load consisting of a 75-Ω resistor in series with a capacitor of unknown value. It was found that at 10 MHz, the voltage standing wave ratio on the line was 3. Find (a) the value of the capacitor and (b) the magnitude of the reflection coefficient.

A 1.05-GHz generator with impedance Z_g =10 Ohms and voltage signal given by,

$$V_g(t) = 10\cos(\omega t - 60°) \quad \text{(V)}$$

is connected to a load $Z_L = 100 + j50$ Ohms through a 50-Ohm, 67-cm lossless transmission line. The wave velocity along the line is $0.7c$. Find,

a) The input impedance at the source side.
b) The voltage reflection coefficient at the load side.
c) The voltage standing wave ratio.

A 50-Ω lossless transmission line is connected to a voltage source with V_g = 10 $\angle 0$ (V) and $Z_g = 50 - j40$ Ohms and is terminated with a load of $j40$ Ohms. If the line is 100 m long with β= 0.25 rad/m, calculate (a) Z_{in} at the source side (b) V at 30 m from the source side (c) the reflection coefficient at the load side.

Consider a distortion-less line with R'= 20 Ω/m, L'= 0.45 μH/m, and C'= 82 pF/m operating at 250 MHz.

(a) Determine the propagation constant, wave speed along the line and the characteristic impedance.
(b) How far will a voltage wave travel before it is reduced to 25% of its initial magnitude?
(c) How far will it travel to suffer a 45° phase shift?

Show that, for a shorted lossless line with length $\lambda/4$, $Z_{in} = j\infty$

Consider a dielectric-filled rectangular cavity with dimensions a=5 cm, b=3 cm and c=3 cm. One of the TM modes that are excited inside this cavity has x-component magnetic field given by

$$H_x(x, y, z, t) = 36\sin(40\pi x)\cdot\cos(100\pi y)\cdot\sin(6.4\pi \times 10^9\, t) \qquad \text{(mA/m)}$$

Find (a) the number of this mode, (b) the dielectric constant of the cavity filling material and (c) all the TE modes that are excited before the given mode.

You are required to select between two filling dielectric materials inside a rectangular metallic waveguide, depending on which one offers the largest frequency range for single-TM-mode operation.
The parameters of the waveguide are given below.

$$a = 2\ cm \quad b = 4\ cm$$

The filling materials are:

Material 1: $\varepsilon_r = 2.2,\ \mu_r = 1$ Material 2: $\varepsilon_r = 3.6,\ \mu_r = 1$

(a) Which filling material would you select? Show your work and calculations.
(b) For the case you select, calculate the resonance frequency of the dominant mode.

It is required to design an air-filled metallic cavity that has a cubical structure ($a = b = c$) and a dominant mode at 2 GHz. (a) Find the required cavity size (b) Give the z-component of the magnetic field of the dominant mode. (c) Find the frequency range of single-mode operation.

Design an air-filled cubic cavity such that it has a dominant TE mode at 12.6 GHz. If the cavity walls are made of copper, find the cavity bandwidth $\left(\Delta f \cong \frac{f_{mnp}}{Q_{mnp}}\right)$. The skin depth of copper at this frequency is 5.9×10-7 m.

(a) Consider a 1-m-long car radio antenna operating in the AM frequency of 1.5 MHz. How much current is required to transmit 4 W of power? (b) If the maximum power density radiated by a half-wave dipole is 50 (mW/m2) at a range of 1 km, what is the current amplitude I_o?

A 50-cm long antenna is excited from its middle point by a sinusoidal current of amplitude I_o=5 A and frequency f. Find the total power radiated and the radiation resistance if the frequency f is (a) 1 MHz and (b) 300 MHz. Also, calculate the maximum magnetic field magnitude 2 km away from the antenna, in each case.
